

Q-Regularized Generative Auto-Bidding: From Suboptimal Trajectories to Optimal Policies

Mingming Zhang*

mingmingzhang@whu.edu.cn
Key Laboratory of Aerospace
Information Security and Trusted
Computing, Ministry of Education,
School of Cyber Science and
Engineering, Wuhan University
Taobao & Tmall Group of Alibaba
Wuhan, China

Na Li*

wannal@whu.edu.cn
Key Laboratory of Aerospace
Information Security and Trusted
Computing, Ministry of Education,
School of Cyber Science and
Engineering, Wuhan University
Taobao & Tmall Group of Alibaba
Wuhan, China

Feiqing Zhuang

Taobao & Tmall Group of Alibaba
Hangzhou, China
zhuangfeiqing.zfq@alibaba-inc.com

Hongyang Zheng

Taobao & Tmall Group of Alibaba
Hangzhou, China
zhenghongyang.hy@alibaba-inc.com

Jiangbing Zhou

Taobao & Tmall Group of Alibaba
Hangzhou, China
zhoujiangbing.zjb@alibaba-inc.com

Wuyin Wang

Taobao & Tmall Group of Alibaba
Hangzhou, China
wangwuyin.wwy@alibaba-inc.com

Shengjie Sun

Taobao & Tmall Group of Alibaba
Hangzhou, China
shengjie.ssj@alibaba-inc.com

Xiaowei Chen

Taobao & Tmall Group of Alibaba
Hangzhou, China
qisheng.cwx@alibaba-inc.com

Junxiong Zhu

Taobao & Tmall Group of Alibaba
Hangzhou, China
xike.zjx@alibaba-inc.com

Lixin Zou

zoulixin@whu.edu.cn
Key Laboratory of Aerospace
Information Security and Trusted
Computing, Ministry of Education,
School of Cyber Science and
Engineering, Wuhan University
Wuhan, China

Chenliang Li[†]

cllee@whu.edu.cn
Key Laboratory of Aerospace
Information Security and Trusted
Computing, Ministry of Education,
School of Cyber Science and
Engineering, Wuhan University
Wuhan, China

Abstract

With the rapid development of e-commerce, auto-bidding has become a key asset in optimizing advertising performance under diverse advertiser environments. The current approaches focus on reinforcement learning (RL) and generative models. These efforts imitate offline historical behaviors by utilizing a complex structure with expensive hyperparameter tuning. The suboptimal trajectories further exacerbate the difficulty of policy learning.

To address these challenges, we propose QGA, a novel Q-value regularized Generative Auto-bidding method. In QGA, we propose to plug a Q-value regularization with double Q-learning strategy into the Decision Transformer backbone. This design enables joint

optimization of policy imitation and action-value maximization, allowing the learned bidding policy to both leverage experience from the dataset and alleviate the adverse impact of the suboptimal trajectories. Furthermore, to safely explore the policy space beyond the data distribution, we propose a Q-value guided dual-exploration mechanism, in which the DT model is conditioned on multiple return-to-go targets and locally perturbed actions. This entire exploration process is dynamically guided by the aforementioned Q-value module, which provides principled evaluation for each candidate action. Experiments on public benchmarks and simulation environments demonstrate that QGA consistently achieves superior or highly competitive results compared to existing alternatives. Notably, in large-scale real-world A/B testing, QGA achieves a 3.27% increase in Ad GMV and a 2.49% improvement in Ad ROI¹.

*Both authors contributed equally to this research. Work done when Mingming Zhang and Na Li were interns at Taobao & Tmall Group of Alibaba.

[†]Corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License.
KDD '26, Jeju Island, Republic of Korea

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2258-5/2026/08

<https://doi.org/10.1145/3770854.3783950>

CCS Concepts

• Information systems → Computational advertising; • Applied computing → Online shopping.

¹<https://github.com/M2C-Tech/QGA>

Keywords

Auto Bidding, Offline Reinforcement Learning, Generative Decision Model

ACM Reference Format:

Mingming Zhang, Na Li, Feiqing Zhuang, Hongyang Zheng, Jiangbing Zhou, Wuyin Wang, Shengjie Sun, Xiaowei Chen, Junxiong Zhu, Lixin Zou, and Chenliang Li. 2026. Q-Regularized Generative Auto-Bidding: From Suboptimal Trajectories to Optimal Policies. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.1 (KDD '26)*, August 09–13, 2026, Jeju Island, Republic of Korea. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3770854.3783950>

KDD Availability Link:

The source code of this paper has been made publicly available at <https://doi.org/10.5281/zenodo.18030706>.

1 Introduction

Superior bidding strategies can bring better advertising performance and increase advertisers' returns [28, 33, 34, 49]. In modern advertising systems, the sheer volume of traffic, the rapid pace of market fluctuations, and the diverse preferences and demands of advertisers together create an extremely complex decision-making environment [1, 21, 23]. To meet these demands, bidding systems must be able to process a massive number of ad impressions in real time and provide reasonable bid prices under diverse requirements [7, 29, 42, 48]. This renders traditional, manually-operated bidding methods inadequate for modern needs [4, 43].

Generative decision models, such as Decision Diffusion (DD) [16] and Decision Transformer (DT) [8], effectively utilize historical behaviors for decision-making, have emerged as promising approaches for auto-bidding. However, these vanilla generative models highly rely on imitating offline behavioral policies over suboptimal behavioral trajectories. Figure 1 shows the MAPE [10] and performance scores of different models on the offline dataset [39]. The behavior cloning (BC) method [2] with a shallow DNN has the highest MAPE and performs worst, while DT achieves the lowest MAPE but only moderate performance. Our method, with moderate MAPE, achieves the best performance instead, indicating that properly deviating from the behavior policy with exploration enhances results.

In recent years, generative bidding models equipped with exploration capabilities have demonstrated impressive performance [15, 16, 22, 30, 31]. GAS [31] designed a post-training search mechanism based on Monte Carlo Tree Search [5, 40] and a value critic to expand the action space. GAVE [15], on the other hand, proposed a framework for value-guided optimization of DT itself. However, these methods also have notable limitations. GAS's optimization only occurs during post-training search, and the process remains confined to a limited set of candidate actions. Furthermore, the backbone model (DT) essentially remains constrained by the quality of the behavioral policy, hindering true global optimization and strong generalization. GAVE involves multiple learning loss functions and complex action exploration mechanisms for policy optimization, requiring extensive hyperparameter tuning. This restricts its capacity to cope with complex and dynamic advertising environments [16, 18], and its ability for out-of-distribution (OOD) policy exploration remains limited.

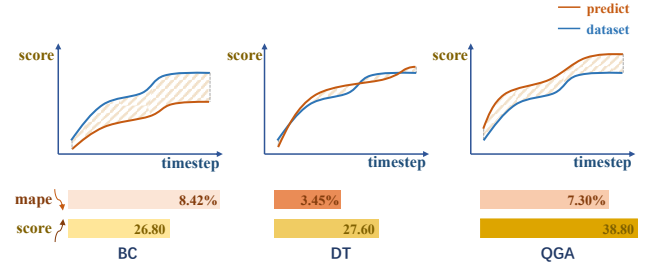


Figure 1: MAPE and performance scores of different models. Lower MAPE indicates stronger, while higher MAPE indicates poorer model imitation ability.

To address these obstacles, we propose a novel Q-regularized generative auto-bidding method with a more compact yet robust model architecture. Specifically, our approach introduces a Q-value regularization term into DT, enabling joint optimization of policy imitation and action value maximization. In this way, we can both leverage experience from the dataset and move beyond the constraints imposed by suboptimal trajectories, favoring better actions. Furthermore, to ensure safe OOD exploration, we design a dual policy exploration mechanism based on the Q-value module. By conditioning DT on a set of perturbed return-to-go (RTG) values, and perturbing multiple candidate actions jointly, we enable a dual search in both the RTG space and the local action neighborhood. The learned Q-value module can efficiently evaluate candidate actions and assist in selecting the optimal one, thereby facilitating the discovery of better bidding strategies in real-world advertising systems.

We conduct comprehensive empirical evaluations of the proposed method using both publicly available offline datasets and simulated advertising environments. Additionally, we validate its practical effectiveness through online A/B testing within large-scale production systems. The experimental results demonstrate strong and robust improvements over both traditional reinforcement learning (RL) methods and state-of-the-art generative approaches, particularly on the challenging AuctionNet-Sparse dataset, as measured by key business metrics.

In summary, our principal contributions are as follows:

- We propose a novel Q-regularized generative auto-bidding method that simultaneously enables effective policy imitation and adaptive policy improvement, providing strong support for auto-bidding under diverse advertiser demands.
- We further propose a dual policy exploration mechanism for better OOD exploration. This cross search driven by the multi-RTG conditioning with action perturbation can help us go beyond the suboptimal datasets for superior bidding strategies.
- We provide comprehensive experimental evaluation using public offline datasets, simulation environments and large-scale real-world advertising platforms. The results demonstrate significant and robust improvements over both state-of-the-art generative and mainstream bidding algorithms, particularly on the challenging AuctionNet-Sparse dataset, across key business metrics.

2 Related Works

Advertisement Auto-bidding. The development of automated bidding techniques, a cornerstone of modern advertising auctions, can be divided into four stages. Early methods used Proportional-Integral-Derivative (PID) controllers and their variants [9, 45, 50], optimizing budget consumption as a proxy for performance, but proved inadequate for complex scenarios. Subsequently, rule-based algorithms [46, 47] emerged, forecasting auction traffic and directly optimizing objectives through parameter selection. The third stage introduced reinforcement learning (RL) approaches, ranging from classical RL [6, 44, 51] to offline [18] and online RL methods [37]. Although these RL methods have achieved good results, they do not perform well in modeling long-term trajectories under complex scenarios. The recent automatic bidding solutions are based on generative decision models. DiffBid is a novel generative advertising bidding model based on the Decision Diffusion [16]. Subsequently, several models leveraging the Decision Transformer, such as GAVE [15] and GAS [31], have been introduced. However, GAVE suffers from high structural complexity and its performance heavily relies on extensive hyperparameter tuning, while GAS depends on post-training search over the limited candidate actions.

Reinforcement Learning and Generative Decision Making. RL has achieved significant success in sequential decision-making tasks [26, 36, 38]. Offline RL enables policy learning from static datasets, which is valuable in high-risk or high-cost domains. To address distribution shift and value overestimation issues in offline RL, approaches like BCQ [14], CQL [25], and IQL [24] have been developed. Nevertheless, traditional methods are limited by their reliance on Markov Decision Processes (MDP), restricting their ability to model long-term dependencies.

Recently, generative decision-making approaches have emerged by framing RL as a sequence modeling problem. The Decision Transformer (DT) [8] utilizes transformer architectures to capture temporal and historical patterns, achieving state-of-the-art results in offline RL. Further extensions, such as the Conditional Decision Transformer (CDT) [35], enable constraint adaptation and improved generalization. QT [20] leverages a Q-value guidance mechanism to enhance the ability of DT to learn better actions. Inspired by this line, our work exploits a Q-value regularization based DT framework and design a novel dual policy exploration mechanism for exploring OOD actions.

3 Preliminary

3.1 Problem Formulation: Auto-Bidding

3.1.1 Auction Environment. We consider an auction environment in which a sequence of I impression opportunities arrives sequentially at discrete time steps $i = 1, \dots, I$. At each time step i , advertisers participate by submitting bids denoted as $\{b_i\}$. An advertiser wins the i -th impression if their bid b_i exceeds the highest bid submitted by other competitors, denoted as b_i^- . Under the generalized second-price (GSP) auction mechanism [11], the winning advertiser pays a cost c_i , which is equal to the second-highest bid. For each impression won, the advertiser receives a private value v_i , often determined by metrics such as the predicted conversion probability or click-through rate.

3.1.2 Optimization Objective and Constraints. During an advertising period, the advertiser's primary objective is to maximize the total value obtained from winning impression opportunities, subject to both budget and performance constraints. Let $x_i \in \{0, 1\}$ indicate whether impression i is won ($x_i = 1$) or not ($x_i = 0$). The optimization problem can be formalized as follows:

$$\begin{aligned} \max_{x_i} \quad & \sum_{i=1}^I x_i v_i \\ \text{s.t.} \quad & \sum_{i=1}^I x_i c_i \leq B \\ & \frac{\sum_{i=1}^I x_i c_i}{\sum_{i=1}^I x_i v_i} \leq C \\ & x_i \in \{0, 1\}, \quad \forall i \end{aligned} \quad (1)$$

where v_i denotes the value of impression i , c_i is the corresponding cost, $B \in \mathbb{R}^+$ represents the total budget, and $C \in \mathbb{R}^+$ specifies the maximum allowable cost per acquisition (CPA).

This formulation ensures that the advertiser neither exceeds the allocated budget nor surpasses the predefined upper limit for average cost per acquisition. Such a multi-constraint framework is crucial for balancing cost efficiency and advertising effectiveness.

3.1.3 Analytical Bidding Solution. Previous research [18] reformulated the auto-bidding problem as a linear programming (LP) problem, allowing for an analytical solution to the optimal bidding policy. In the case of a single CPA constraint, the optimal bid simplifies to:

$$b_i^* = \lambda^* v_i, \quad \text{where} \quad \lambda^* = \lambda_0^* + \lambda_1^* C \quad (2)$$

This reformulation reduces the control problem to dynamically learning or adjusting the unified bidding parameter λ^* to optimize performance while satisfying all constraints [18, 39].

3.2 Sequential Decision Modeling

In the rapidly evolving field of online advertising, bidding policies must dynamically adapt to optimize campaign performance. Traditional static or myopic approaches often fail to account for auction volatility, fluctuating budget constraints, and time-sensitive objectives. Drawing inspiration from generative bidding frameworks, we thus conceptualize auto-bidding as a *sequential decision-making* problem, wherein each decision influences outcomes throughout the entire campaign horizon.

Formalization as a Markov Decision Process. We model auto-bidding as a sequential decision process in which an agent interacts with the advertising environment E at discrete time steps. Specifically, at each time step t : 1) The agent observes the current **state** $s_t \in \mathcal{S}$, which encapsulates relevant contextual information; 2) Based on s_t , the agent selects an **action** $a_t \in \mathcal{A}$, representing the bid amount; 3) The environment transitions according to underlying, potentially unknown, dynamics $\mathcal{T}$. In the Markov Decision Process (MDP) framework, the transition function $\mathcal{T} : s_t \times a_t \rightarrow s_{t+1}$ indicates that the next state s_{t+1} is determined solely by the current state and action. On contrary, the transition for auto-bidding may depend on the full trajectory τ ; 4) The agent receives a **reward** r_t , quantifying the value obtained at time t . This process is repeated

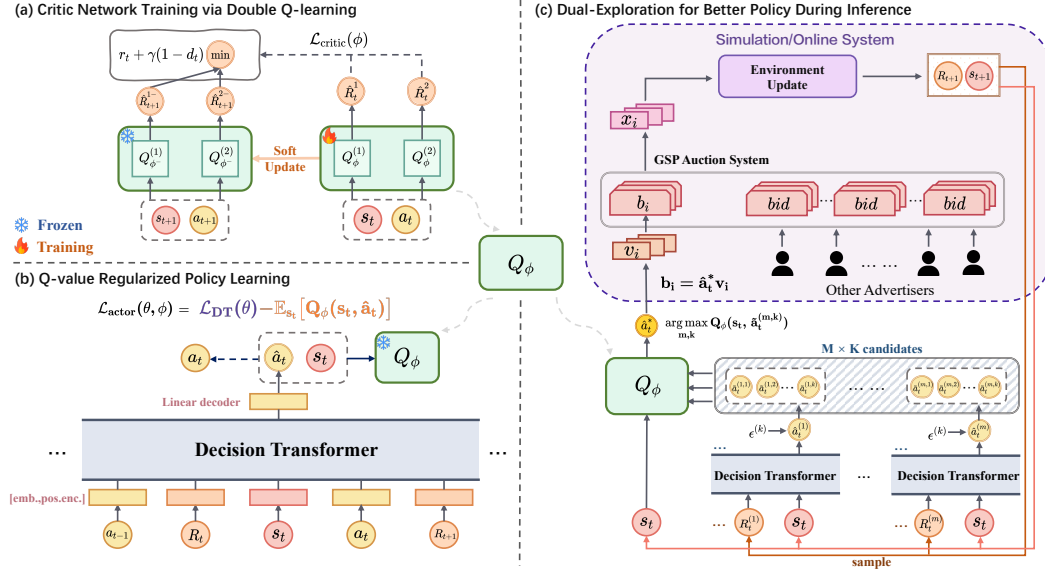


Figure 2: Overall structure of QGA.

until the end of the bidding period (e.g., a day), with the primary objective of maximizing the cumulative reward.

Modeling Auto-Bidding. According to the above setting, we further refine the auto-bidding problem as follows:

- **State s_t :** A feature vector representing the contextual state at time t , including remaining campaign duration, available budget, and historical bidding statistics.
- **Action a_t :** The bidding parameter selected at time t , denoted as $a_t = \lambda_t$, consistent with the optimal bidding policy (cf. Eq. (3)).
- **Reward r_t :** Given N_t impressions delivered between t and $t + 1$, the reward is defined as $r_t = \sum_{n=t}^{N_t} x_{nt} v_{nt}$, representing the total value accrued during that interval.
- **Return-to-Go R_t :** The cumulative anticipated reward from time t onward, defined as $R_t = \sum_{t'=t}^T r_{t'}$, where T denotes the campaign horizon.

4 Method

In this section, we first present a detailed explanation of constructing DT as the fundamental backbone of the bidding model. Then, we describe the incorporation of Q-value regularization into policy learning. Finally, we discuss how to leverage the Q-value regularized model on the existing dataset to explore policies that extend beyond the current behavior policy.

4.1 DT for Bidding Policy Learning

Here, we employ DT [8] as the backbone model for learning sequential bidding policies. DT is a trajectory-based model that formulates sequential decision-making as a conditional sequence modeling problem, leveraging the powerful representation learning capabilities of Transformer architectures.

4.1.1 Trajectory Formulation. To adapt DT for the auto-bidding task, we first represent the agent’s experience as a collection of trajectories. Each trajectory τ consists of a sequence of (R_t, s_t, a_t) tuples. During training, the DT model learns to predict the optimal next action a_t conditioned on the desired future return R_t and the observed state s_t . This sequence modeling approach enables DT to learn the bidding policy of the existing trajectory.

4.1.2 Sequence Modeling with Decision Transformers. The Decision Transformer uses causal Transformer blocks to process the sequential input tokens. At each time step t , the model receives embeddings of (R_t, s_t, a_t) . In practice, the input to DT consists of the advertising bidding trajectory from the most recent K steps.

$$(R_{t-K+1}, s_{t-K+1}, a_{t-K+1}, R_{t-K+2}, s_{t-K+2}, a_{t-K+2}, \dots, R_t, s_t) \quad (3)$$

These embeddings are processed by the Decision Transformer to produce contextualized representations, from which the model predicts the next action a_{t+1} based on the history up to time t and the target return-to-go R_{t+1} .

4.1.3 Training Objective. The model is trained via a supervised learning objective: given a dataset of historical bidding trajectories, the Decision Transformer is optimized to minimize the mean squared error (MSE) between the predicted action $\hat{a}_t$ and the ground-truth action a_t taken in the observed trajectory:

$$\mathcal{L}_{DT} = \mathbb{E}_{\tau \sim \mathcal{D}} \mathbb{E}_{t_0 \sim \mathcal{U}[1, T-K+1]} \left[\frac{1}{K} \sum_{k=0}^{K-1} \mathbb{E} \left[(\hat{a}_{t_0+k} - a_{t_0+k})^2 \mid (R_{t_0:t_0+K}, s_{t_0:t_0+K}, a_{t_0:t_0+K-1}) \right] \right] \quad (4)$$

By conditioning on various values of R_t during training, the Decision Transformer can generate the action with respect to different levels of campaign performance, which supports controllable trade-offs between value maximization and constraint satisfaction.

4.2 Q-value Regularized Policy Learning

Although the Decision Transformer framework enables behavioral cloning from offline trajectories, its performance is fundamentally limited by the quality of the behavior policy in the dataset. The standard Decision Transformer only learns to imitate observed actions, lacking any explicit mechanism to favor actions that could lead to higher returns. To address this limitation, inspired by [20], we introduce a modification that allows the policy to utilize value estimates from an auxiliary critic network. This enhancement enables the generation of trajectories with higher expected returns than imitation alone.

4.2.1 Actor Loss with Q-value Regularization. Let s_t and a_t denote the state and action at time t , and let $\hat{a}_t$ represent the action predicted by the Decision Transformer given the current historical context. To encourage the Decision Transformer to predict high-return actions, we introduce a critic network $Q_\phi(s_t, a_t)$, parameterized by ϕ , which estimates the expected return for each state-action pair. The regularization term for the actor loss is defined as:

$$\mathcal{L}_Q(\phi) = -\mathbb{E}_{s_t} [Q_\phi(s_t, \hat{a}_t)] \quad (5)$$

Combining this with the original Decision Transformer loss, we define the overall actor loss as:

$$\mathcal{L}_{\text{actor}}(\theta, \phi) = \mathcal{L}_{\text{DT}}(\theta) + \alpha \cdot \mathcal{L}_Q(\phi) \quad (6)$$

where θ are the parameters of the Decision Transformer, and $\alpha > 0$ is a regularization coefficient. In this way, the Decision Transformer is trained both to imitate actions from the dataset and to maximize the predicted return according to the learned critic.

4.2.2 Critic Training. The critic network serves as an estimator for the action-value function $Q_\phi(s_t, a_t)$, which is essential for providing value-based feedback to the actor. In accordance with the principles of double Q-learning[17, 41], we maintain two distinct Q-networks, $Q_\phi^{(1)}$ and $Q_\phi^{(2)}$, with the same parameters ϕ . During training, for each transition in the batch, we obtain $(s_t, a_t, r_t, s_{t+1}, d_t)$, where r_t is the observed reward and d_t is a binary indicator for episode termination.

To ensure stable value estimation, the target Q-value is calculated using a separate target network, whose parameters, ϕ^- , are an exponential moving average of the main critic's parameters[13, 32]. At each step, we compute the predicted next action $\hat{a}_{t+1}$ as the actor's output for state s_{t+1} in the sequence. The Bellman target [3] for each transition is then given by:

$$y_t = r_t + \gamma(1 - d_t) \cdot \min_{i \in \{1, 2\}} Q_{\phi^-}^{(i)}(s_{t+1}, a_{t+1}) \quad (7)$$

where γ denotes the discount factor. By taking the minimum of the two target Q-values, we mitigate positive bias and obtain more conservative and reliable targets.

The critic loss aggregates the squared temporal-difference (TD) errors for both Q-heads over the batch:

$$\mathcal{L}_{\text{critic}}(\phi) = \frac{1}{N} \sum_{j=1}^N \left[\left(Q_\phi^{(1)}(s_{t_j}, a_{t_j}) - y_{t_j} \right)^2 + \left(Q_\phi^{(2)}(s_{t_j}, a_{t_j}) - y_{t_j} \right)^2 \right] \quad (8)$$

where N denotes the number of valid non-terminal positions within the batch.

After computing gradients of $\mathcal{L}_{\text{critic}}$ with respect to ϕ , we update the critic network parameters via gradient descent. To promote temporal consistency and smoothness of the targets, the target network parameters are updated at each step via exponential moving averaging:

$$\phi^- \leftarrow (1 - \tau)\phi^- + \tau\phi \quad (9)$$

where τ is a small positive value.

The critic training method enables the Q-network to accurately estimate values based on offline trajectories, thereby providing essential support to QGA. During training, Q-value regularization by the critic not only guides the actor to imitate the dataset policy but also encourages the maximization of action values, thereby integrating value-based objectives into the learning process. Furthermore, within the policy exploration mechanism, the critic functions as an offline value estimator to evaluate and rank diverse candidate actions.

4.3 Dual-Exploration for Better Policy

Although Q-value regularization encourages the Decision Transformer to select actions with higher estimated value within the offline dataset, the resulting policy is still fundamentally constrained by fragments of the behavior trajectories[19, 27]. No matter training or inference, the actions generated by the Decision Transformer typically stay close to those in the historical trajectories or are stitching of fragments from these trajectories. This constraint limits its capacity to explore and identify novel or potentially superior bidding policies.

4.3.1 Q-value based Policy Improvement. To further alleviate the limitations of behavior cloning and Q-value regularization, we introduce a policy improvement strategy during inference by combining **multi-return-to-go (multi-rtg) conditioning** with **action perturbation**. This approach enables broader and safer off-policy exploration, substantially enhancing the agent's ability to identify high-value actions even outside the empirical support of the offline dataset.

Concretely, for each decision step t , our method proceeds as follows:

- (1) **Multi-RTG Sampling:** We sample a set of M different candidate return-to-go values $R_t^{(1)}, \dots, R_t^{(M)}$, usually centered around the current estimated return. Each return-to-go represents a hypothetical future performance target for the Decision Transformer model.
- (2) **Action Generation:** For each candidate $R_t^{(m)}$, the Decision Transformer predicts an action $\hat{a}_t^{(m)}$ conditioned on $R_t^{(m)}$ and the episode history.
- (3) **Action Perturbation:** To further diversify the action space, we add small random noise to each predicted action to obtain K perturbed actions, i.e., $\tilde{a}_t^{(m,k)} = \hat{a}_t^{(m)} + \epsilon^{(k)}$, where $\epsilon^{(k)}$ is drawn from Gaussian noise distribution.
- (4) **Q-value Evaluation and Selection:** All candidates (across different $R_t^{(m)}$ and perturbations k) are evaluated by the learned critic network to obtain Q-value estimates $q^{(m,k)} = Q_\phi(s_t, \tilde{a}_t^{(m,k)})$. The final deployed action is the one with the

highest estimated Q-value:

$$\hat{a}_t^* = \arg \max_{m,k} Q_\phi(s_t, \hat{a}_t^{(m,k)}). \quad (10)$$

This process effectively constructs a two-dimensional search over possible future objectives (through multi-rtg conditioning) and local action variations (via perturbation), all guided by the offline-trained value function. It enables a significantly richer and more adaptive exploration of the action space compared to naive imitation or single-rtg Q-regularized methods. Our policy improvement approach provides a principled and practical strategy for substantially increasing the likelihood of discovering novel actions with superior long-term value in auto-bidding.

5 Experiments

In this section, we present extensive evaluations of the proposed QGA, including both offline testing and testing in a simulation environment. Comprehensive ablation studies and parameter experiments are also carried out to address the following questions:

- RQ1: How does QGA perform compared to other traditional and generative baseline methods?
- RQ2: Can QGA adapt to more complex and realistic environments?
- RQ3: How do the proposed core components contribute to the performance of QGA?
- RQ4: How do different parameter settings affect the performance?

5.1 Datasets

To ensure fairness and reproducibility, we evaluated on AuctionNet [39], a large-scale, publicly available benchmark released by Alimama and widely used for advertising auto-bidding research.

Specifically, we used two datasets: *AuctionNet* and *AuctionNet-Sparse*, corresponding to dense and sparse conversion scenarios, respectively. Each dataset contains about 480,000 bidding trajectories, with 48 time steps in each trajectory. In addition, we utilized the AuctionNet simulation environment for realistic evaluation. More dataset details are provided in Appendix A.1.

5.2 Evaluation Settings

5.2.1 Metrics. We evaluated model effectiveness using the advertising bidding score, which reflects the agent’s ability to maximize conversions while adhering to the advertiser’s CPA constraint C . If the actual CPA exceeds C , a penalty is imposed. The score is defined as:

$$\text{Score} = \mathbb{P}(\text{CPA}; C) \cdot \sum_i x_i \cdot V_i \quad (11)$$

the penalty function for exceeding the CPA constraint is given by:

$$\mathbb{P}(\text{CPA}; C) = \min \left\{ \left(\frac{C}{\text{CPA}} \right)^\beta, 1 \right\} \quad (12)$$

with $\beta > 0$ (typically, $\beta = 2$). The penalty is only enforced when $\text{CPA} > C$.

5.2.2 Evaluation Protocol. We followed the methodology of AuctionNet [39]. During offline evaluation, a bid is considered successful if it exceeds the historical minimum winning price recorded in the test dataset. To comprehensively evaluate the performance of one specific model across different advertisers, we sequentially used it to act as the agent for each advertiser while the remaining advertisers served as competitors. The agent’s final score is computed using Equation 11 at the end of the delivery period.

AuctionNet also provides a standardized simulated online environment, where ad delivery consists of 48 decision time steps. At each step, multiple advertisers bid for numerous impression opportunities. Further details on the simulation environment are presented in Appendix A.2.

5.2.3 Baselines. We evaluate QGA on the auto-bidding task against several representative baselines covering offline reinforcement learning and generative modeling paradigms. For offline reinforcement learning, we include **BC** [2], **IQL** [24], **CQL** [25] and **TD3-BC** [12]. For generative methods, we consider **DiffBid** [16], which is based on Decision Diffusion, as well as **DT** [8] and its variants, **GAS** [31] and **GAVE** [15].

5.2.4 Deployment Settings. All traditional reinforcement learning baselines [12, 24, 25] were implemented using the architectures and parameter settings specified in the AuctionNet benchmark [39]. DiffBid [16] was implemented according to the original paper’s details and settings. For GAVE [15], we reuse the original paper’s performance as we couldn’t reproduce its results.

5.3 RQ1: Evaluation Results of Offline Testing

Table 1 reports offline test results for QGA and multiple baselines on both AuctionNet and AuctionNet-Sparse datasets under different budget constraints. Our key findings are as follows:

Consistently Superior Performance: QGA consistently achieves either the best or highly competitive results across all budget levels on both datasets. Notably, on the more challenging AuctionNet-Sparse dataset, QGA exhibits pronounced superiority, particularly as the budget increases. The Sparse dataset, with its sparse conversion rates, more closely resembles real-world scenarios. These results demonstrate QGA’s strong adaptability and generalization in offline bidding environments—especially under sparse conversion circumstances—thus providing better alignment with practical, real-world settings.

Comparison with DT-based Methods: While Decision Transformer and its variants (e.g., GAS, GAVE) outperform traditional RL baselines due to their advanced sequence modeling abilities, QGA consistently matches or exceeds the performance of these methods, particularly at higher budget levels. The performance improvements observed on the AuctionNet dataset are less substantial than those on the AuctionNet-Sparse dataset, which may be attributed to the much higher conversion rate in AuctionNet (Appendix A.1), thereby diminishing the relative impact of policy optimization.

Imitation behavior cannot produce better policies: BC and DT, which rely solely on imitation, underperform compared to methods with exploration (e.g., IQL, GAS, GAVE). This indicates

Table 1: Offline evaluation on the AuctionNet and AuctionNet-Sparse datasets. Bold indicates the best results, underline denotes the second-best results, and "*" marks results obtained directly from the original paper.

Dataset	Budget	IQL	BC	CQL	TD3-BC	DT	DiffBid	GAS*	GAVE*	QGA
AuctionNet [39]	50%	165	169	77	167	170	72	193	<u>201</u>	206
	75%	234	236	116	239	232	129	<u>287</u>	296	282
	100%	293	285	158	288	298	169	<u>359</u>	376	353
	125%	331	342	194	332	340	221	<u>409</u>	421	<u>409</u>
	150%	378	375	231	376	379	280	461	467	<u>463</u>
AuctionNet-Sparse [39]	50%	17.9	15.0	16.1	15.0	18.4	10.7	<u>18.4</u>	19.6	19.6
	75%	26.9	20.3	22.4	22.7	24.9	22.2	27.5	<u>28.3</u>	29.0
	100%	30.9	26.8	27.9	26.4	27.6	24.6	36.1	<u>37.2</u>	38.8
	125%	32.0	31.6	32.1	31.4	35.6	31.8	40.0	<u>42.7</u>	45.3
	150%	37.8	36.6	37.6	38.0	39.4	36.5	46.5	<u>47.4</u>	50.1

that imitation learning from suboptimal data is insufficient, and exploration mechanisms are crucial for better performance.

Suboptimal Performance of DiffBid: DiffBid underperforms on both datasets, likely due to sparse rewards and dynamic, long-horizon environments. Limited reward signals impede learning, and changing dynamics undermine effective reverse process inference, resulting in poor trajectory and action quality.

5.4 RQ2: Evaluation Results in Simulation Environment

To evaluate the bidding capabilities of QGA in complex and realistic auction scenarios, we conducted experiments in simulated environments. As demonstrated by the results in Table 2, QGA achieves a score of 8113 in the simulation environment, substantially outperforming all baseline methods, including IQL (6534), CQL (7138), and GAS (7454). This significant performance improvement indicates that QGA is highly effective at adapting to the complexities of the simulation environment. These results provide strong empirical evidence that QGA can generalize well and maintain high performance in complex, dynamic simulated settings.

5.5 RQ3: Ablation Study

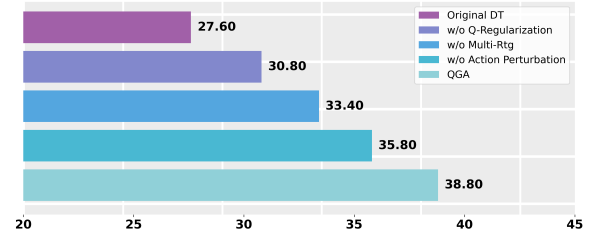
Figure 3 presents the results of the ablation study. QGA, with the full configuration, achieves the highest score of 38.8. When either the Action Perturbation or Multi-RTG module is removed, the scores decrease to 35.8 and 33.4, respectively. Removing Q-Regularization leads to the largest drop in performance, resulting in a score of 30.8. In contrast, the original baseline DT model attains only 27.6. These findings demonstrate that each proposed module independently enhances model performance, and their combination yields complementary effects that further improve overall results. This ablation analysis confirms the necessity and synergy of both the Q-Regularization and Dual-Exploration module in improving the effectiveness of the model.

5.6 RQ4: Hyper-parameter Study

We conducted a comprehensive hyper-parameter study to investigate the effects of the regularization coefficient α in the actor loss, as well as the number of return-to-go samples (M) and action samples (N) utilized in the dual-exploration module.

Table 2: Performance in the Simulation Environment

Dataset	Method	Score
Simulation Env.	IQL	6534
	BC	6366
	CQL	7138
	TD3-BC	7008
	DT	6920
	DiffBid	6248
	GAS	7454
	OURS	8113

**Figure 3: Ablation study**

Effect of Regularization Coefficient α . As defined in Eq. (6), the coefficient α mediates the trade-off between the behavioral cloning loss and Q-value regularization, thus modulating the contribution of the critic network during training. Figure 4a demonstrates that the value of α in the actor loss significantly impacts the final performance. Specifically, we observe that setting $\alpha = 1.0$ yields the highest score, while both lower and higher values result in reduced performance. This finding underscores the importance of appropriately selecting α , with $\alpha = 1.0$ providing the optimal balance between policy regularization and expressiveness.

Effect of the Number of RTG and Action Samples.

Figure 4b shows how varying the number of RTG and action samples in the dual-exploration module affects final performance. Both RTG and Action curves achieve peak scores when the sample number is 3, indicating that moderate stochastic exploration

($M = N = 3$) benefits policy performance. Increasing the number further leads to stable or slightly fluctuating results, possibly because excessive sampling causes actions to deviate too much. These results suggest that using 3 samples achieves a good trade-off between exploration and stability, thus optimizing overall performance.

6 Online A/B Test

To evaluate the practical effectiveness of QGA in real-world advertising systems, we conducted an online A/B test on the Direct Express platform of Taobao, one of the largest e-commerce platforms in China. Experimental setup and results are presented below; more details are in Appendix B.

6.1 Experimental Setup

The online experiment is divided into two phases. We compare the performance of QGA (treatment group) with the baseline DT (control group). The specific setup is as follows:

- **Experiment Duration:**
 - Phase 1: Regular days
 - Phase 2: Promotion period
- **Metrics:**
 - *Ad Click*: Number of clicks on ads
 - *Ad Cost*: Total advertising cost via bidding
 - *Ad GMV*: Gross Merchandise Volume from ad clicks
 - *Ad ROI*: Return on Investment
 - *Global Click*: Total platform clicks (organic + ads)
 - *Global GMV*: GMV from non-ad (organic) clicks
- **Control Strategy:** The control group uses Original DT for different product categories, with both groups aligned on intervention strategies except for the bidding policy.

6.2 Experimental Results

Figure 5 summarizes the key performance metrics for both phases. The following observations can be drawn: 1) **Generalization Across Scenarios**. The method performs well during both the regular period (Phase 1) and the promotion period (Phase 2), demonstrating strong adaptability to fluctuations in traffic volume and changes in user behavior; 2) **Cost Efficiency**. In Phase 1, the experimental group experiences a slight increase in Ad Cost (+1.35%), while Ad GMV rises by 3.27%, resulting in a 2.49% improvement in Ad ROI. In Phase 2, although Ad Cost increases by 2.91%, Ad GMV grows by 4.70%, still achieving a 2.16% uplift in ROI. This indicates flexible cost allocation under varying traffic conditions; 3) **Exploration Benefits**. In Phase 2, with a significant increase in Ad Cost, both Ad GMV and ROI increase noticeably, indicating that the exploration mechanism effectively identifies previously under-exploited bidding strategies, especially in high-traffic scenarios.

The online A/B test confirms that QGA significantly outperforms traditional DT method in real-world advertising systems. It achieves consistent improvements in core metrics while maintaining budget discipline, demonstrating its readiness for full-scale deployment. These results validate the effectiveness of integrating Q-value regularization and dual-exploration mechanism into generative decision models for complex auto-bidding tasks in advertising.

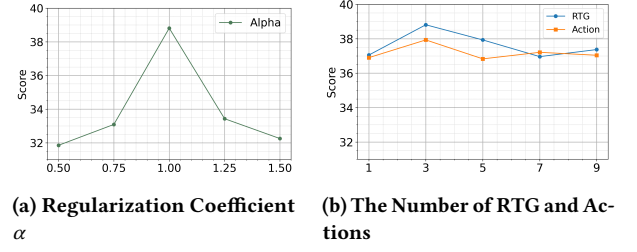


Figure 4: Hyper-parameter Study

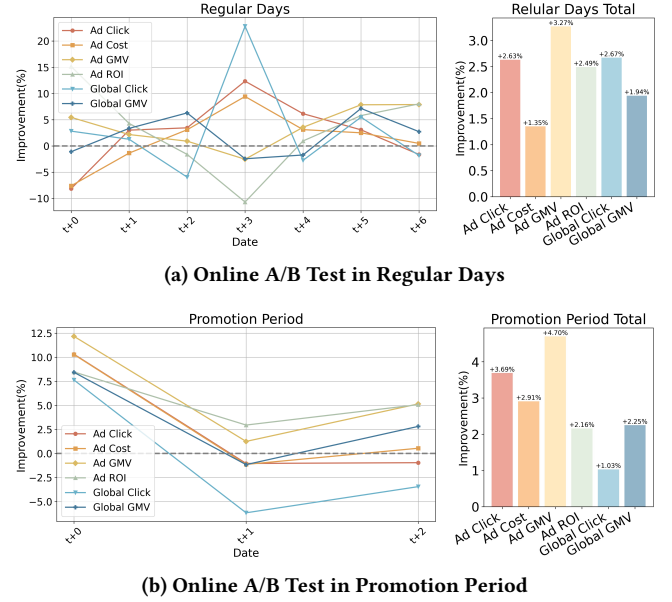


Figure 5: Online A/B Test

7 Conclusion

We propose a Q-value regularized generative auto-bidding method that augments the Decision Transformer with Q-regularization and dual-exploration mechanism. QGA enables the learning of superior bidding policies from offline data and effective exploration of out-of-distribution policies. Extensive experiments on public datasets, simulation environments, and large-scale online A/B tests demonstrate significant improvements over prior methods across key business metrics. QGA offers a practical and robust solution for auto-bidding in real-world advertising systems.

However, QGA has following limitations. In complex environments involving multiple advertisers with unknown competitor bids, the exploration mechanism may not fully capture subtle environmental shifts. Additionally, the model lacks stability measures for abrupt changes in traffic conditions, such as those during promotions. Future work will explore fine-grained control strategies to enhance robustness in these scenarios.

Acknowledgments

This work was supported by Alibaba Group through Alibaba Innovative Research Program; This work was supported by National Natural Science Foundation of China (No. 62272349, No. 62302345, No. U23A20305);

References

- [1] Gagan Aggarwal, Ashwinkumar Badanidiyuru, Santiago R Balseiro, Kshipra Bhawalkar, Yuan Deng, Zhe Feng, Gagan Goel, Christopher Liaw, Haihao Lu, Mohammad Mahdian, et al. 2024. Auto-bidding and auctions in online advertising: A survey. *ACM SIGecom Exchanges* 22, 1 (2024), 159–183.
- [2] Michael Bain and Claude Sammut. 1995. A Framework for Behavioural Cloning. In *Machine intelligence* 15. 103–129.
- [3] Richard Bellman. 1954. The theory of dynamic programming. *Bull. Amer. Math. Soc.* 60, 6 (1954), 503–515.
- [4] Nikolay Borissov, Dirk Neumann, and Christof Weinhardt. 2010. Automated bidding in computational markets: an application in market-based allocation of computing services. *Autonomous Agents and Multi-Agent Systems* 21, 2 (2010), 115–142.
- [5] Cameron B Browne, Edward Powley, Daniel Whitehouse, Simon M Lucas, Peter I Cowling, Philipp Rohlfshagen, Stephen Tavener, Diego Perez, Spyridon Samothakis, and Simon Colton. 2012. A survey of monte carlo tree search methods. *IEEE Transactions on Computational Intelligence and AI in games* 4, 1 (2012), 1–43.
- [6] Han Cai, Kan Ren, Weinan Zhang, Kleanthis Malialis, Jun Wang, Yong Yu, and Defeng Guo. 2017. Real-time bidding by reinforcement learning in display advertising. In *Proceedings of the tenth ACM international conference on web search and data mining*. 661–670.
- [7] Leng Cai, Junxuan He, Yikai Li, Junjie Liang, Yuanping Lin, Ziming Quan, Yawen Zeng, and Jin Xu. 2025. RTBAgent: A LLM-based Agent System for Real-Time Bidding. In *Companion Proceedings of the ACM on Web Conference 2025*. 104–113.
- [8] Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Misha Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. 2021. Decision transformer: Reinforcement learning via sequence modeling. *Advances in neural information processing systems* 34 (2021), 15084–15097.
- [9] Ye Chen, Pavel Berkhin, Bo Anderson, and Nikhil R Devanur. 2011. Real-time bidding algorithms for performance-based display ad allocation. In *Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining*. 1307–1315.
- [10] Arnaud De Myttenaere, Boris Golden, Bénédicte Le Grand, and Fabrice Rossi. 2016. Mean absolute percentage error for regression models. *Neurocomputing* 192 (2016), 38–48.
- [11] Benjamin Edelman, Michael Ostrovsky, and Michael Schwarz. 2007. Internet advertising and the generalized second-price auction: Selling billions of dollars worth of keywords. *American economic review* 97, 1 (2007), 242–259.
- [12] Scott Fujimoto and Shixiang Shane Gu. 2021. A minimalist approach to offline reinforcement learning. *Advances in neural information processing systems* 34 (2021), 20132–20145.
- [13] Scott Fujimoto, Herke Hoof, and David Meger. 2018. Addressing function approximation error in actor-critic methods. In *International conference on machine learning*. PMLR, 1587–1596.
- [14] Scott Fujimoto, David Meger, and Doina Precup. 2019. Off-policy deep reinforcement learning without exploration. In *International conference on machine learning*. PMLR, 2052–2062.
- [15] Jingtong Gao, Yewen Li, Shuai Mao, Peng Jiang, Nan Jiang, Yejing Wang, Qingpeng Cai, Fei Pan, Kun Gai, Bo An, et al. 2025. Generative Auto-Bidding with Value-Guided Explorations. *arXiv preprint arXiv:2504.14587* (2025).
- [16] Jiayan Guo, Yusen Huo, Zhilin Zhang, Tianyu Wang, Chuan Yu, Jian Xu, Bo Zheng, and Yan Zhang. 2024. Generative auto-bidding via conditional diffusion modeling. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 5038–5049.
- [17] Hado Hasselt. 2010. Double Q-learning. *Advances in neural information processing systems* 23 (2010).
- [18] Yue He, Xiujun Chen, Di Wu, Junwei Pan, Qing Tan, Chuan Yu, Jian Xu, and Xiaoqiang Zhu. 2021. A unified solution to constrained bidding in online display advertising. In *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*. 2993–3001.
- [19] Charles A Hepburn and Giovanni Montana. 2022. Model-based trajectory stitching for improved offline reinforcement learning. *arXiv preprint arXiv:2211.11603* (2022).
- [20] Shengchao Hu, Ziqing Fan, Chaoqin Huang, Li Shen, Ya Zhang, Yanfeng Wang, and Dacheng Tao. 2024. Q-value regularized transformer for offline reinforcement learning. *arXiv preprint arXiv:2405.17098* (2024).
- [21] Arti Jha, Harshit Jain, Parikshit Sharma, Yashvardhan Sharma, and Kamlesh Tiwari. 2024. Optimizing Real-Time Bidding Strategies: An Experimental Analysis of Reinforcement Learning and Machine Learning Techniques. *Procedia Computer Science* 235 (2024), 2017–2026.
- [22] Hao Jiang, Yongxiang Tang, Yanxiang Zeng, Pengjia Yuan, Yanhua Cheng, Teng Sha, Xialong Liu, and Peng Jiang. 2025. Optimal Return-to-Go Guided Decision Transformer for Auto-Bidding in Advertisement. In *Companion Proceedings of the ACM on Web Conference 2025*. 1033–1037.
- [23] Dmytro Korenkevych, Frank Cheng, Artsiom Balakir, Alex Nikulkov, Lingnan Gao, Zhihao Cen, Zuobing Xu, and Zheqing Zhu. 2024. Offline reinforcement learning for optimizing production bidding policies. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 5251–5259.
- [24] Ilya Kostrikov, Ashvin Nair, and Sergey Levine. 2021. Offline reinforcement learning with implicit q-learning. *arXiv preprint arXiv:2110.06169* (2021).
- [25] Aviral Kumar, Aurick Zhou, George Tucker, and Sergey Levine. 2020. Conservative q-learning for offline reinforcement learning. *Advances in neural information processing systems* 33 (2020), 1179–1191.
- [26] Sergey Levine, Aviral Kumar, George Tucker, and Justin Fu. 2020. Offline reinforcement learning: Tutorial, review, and perspectives on open problems. *arXiv preprint arXiv:2005.01643* (2020).
- [27] Guanghe Li, Yixiang Shan, Zhengbang Zhu, Ting Long, and Weinan Zhang. 2024. Diffstitch: Boosting offline reinforcement learning with diffusion-based trajectory stitching. *arXiv preprint arXiv:2402.02439* (2024).
- [28] Haoming Li, Yusen Huo, Shuai Dou, Zhenzhe Zheng, Zhilin Zhang, Chuan Yu, Jian Xu, and Fan Wu. 2024. Trajectory-wise iterative reinforcement learning framework for auto-bidding. In *Proceedings of the ACM Web Conference 2024*. 4193–4203.
- [29] Juncheng Li and Pingzhong Tang. 2022. Auto-bidding equilibrium in ROI-constrained online advertising markets. *arXiv preprint arXiv:2210.06107* (2022).
- [30] Kaiyuan Li, Pengyu Wang, Yunshan Peng, Pengjia Yuan, Yanxiang Zeng, Rui Xiang, Yanhua Cheng, Xialong Liu, and Peng Jiang. 2025. EBAReT: Expert-guided Bag Reward Transformer for Auto Bidding. In *Companion Proceedings of the ACM on Web Conference 2025*. 1104–1108.
- [31] Yewen Li, Shuai Mao, Jingtong Gao, Nan Jiang, Yunjian Xu, Qingpeng Cai, Fei Pan, Peng Jiang, and Bo An. 2025. GAS: Generative Auto-bidding with Post-training Search. In *Companion Proceedings of the ACM on Web Conference 2025*. 315–324.
- [32] Timothy P Lillicrap, Jonathan J Hunt, Alexander Pritzel, Nicolas Heess, Tom Erez, Yuval Tassa, David Silver, and Daan Wierstra. 2015. Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971* (2015).
- [33] Mengjuan Liu, Li Jiaxing, Zhengning Hu, Jinyu Liu, and Xuyun Nie. 2020. A dynamic bidding strategy based on model-free reinforcement learning in display advertising. *IEEE Access* 8 (2020), 213587–213601.
- [34] Mengjuan Liu, Wei Yue, Lizhou Qiu, and Jiaxing Li. 2020. An effective budget management framework for real-time bidding in online advertising. *IEEE access* 8 (2020), 131107–131118.
- [35] Zuxin Liu, Zijian Guo, Yihang Yao, Zhepeng Cen, Wenhao Yu, Tingnan Zhang, and Ding Zhao. 2023. Constrained decision transformer for offline safe reinforcement learning. In *International conference on machine learning*. PMLR, 21611–21630.
- [36] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemaire, Alex Graves, Martin Riedmiller, Andreas K Fidjeland, Georg Ostrovski, et al. 2015. Human-level control through deep reinforcement learning. *nature* 518, 7540 (2015), 529–533.
- [37] Zhiyu Mou, Yusen Huo, Rongquan Bai, Mingzhou Xie, Chuan Yu, Jian Xu, and Bo Zheng. 2022. Sustainable online reinforcement learning for auto-bidding. *Advances in Neural Information Processing Systems* 35 (2022), 2651–2663.
- [38] John Schulman, Sergey Levine, Pieter Abbeel, Michael Jordan, and Philipp Moritz. 2015. Trust region policy optimization. In *International conference on machine learning*. PMLR, 1889–1897.
- [39] Kefan Su, Yusen Huo, Zhilin Zhang, Shuai Dou, Chuan Yu, Jian Xu, Zongqing Lu, and Bo Zheng. 2024. Auctionnet: A novel benchmark for decision-making in large-scale games. *Advances in Neural Information Processing Systems* 37 (2024), 94428–94452.
- [40] Maciej Świechowski, Konrad Godlewski, Bartosz Sawicki, and Jacek Mańdziuk. 2023. Monte Carlo tree search: A review of recent modifications and applications. *Artificial Intelligence Review* 56, 3 (2023), 2497–2562.
- [41] Hado Van Hasselt, Arthur Guez, and David Silver. 2016. Deep reinforcement learning with double q-learning. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 30.
- [42] Yuhao Wang, Ha Tsz Lam, Yi Wong, Ziru Liu, Xiangyu Zhao, Yichao Wang, Bo Chen, Huifeng Guo, and Ruiming Tang. 2023. Multi-task deep recommender systems: A survey. *arXiv preprint arXiv:2302.03525* (2023).
- [43] Chao Wen, Miao Xu, Zhilin Zhang, Zhenzhe Zheng, Yuhui Wang, Xiangyu Liu, Yu Rong, Dong Xie, Xiaoyang Tan, Chuan Yu, et al. 2022. A cooperative-competitive multi-agent framework for auto-bidding in online advertising. In *Proceedings of the Fifteenth ACM International Conference on Web Search and Data Mining*. 1129–1139.
- [44] Di Wu, Xiujun Chen, Xun Yang, Hao Wang, Qing Tan, Xiaoxun Zhang, Jian Xu, and Kun Gai. 2018. Budget constrained bidding by model-free reinforcement learning in display advertising. In *Proceedings of the 27th ACM International Conference on Information and Knowledge Management*. 1443–1451.

- [45] Xun Yang, Yasong Li, Hao Wang, Di Wu, Qing Tan, Jian Xu, and Kun Gai. 2019. Bid optimization by multivariable control in display advertising. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*. 1966–1974.
- [46] Hao Yu, Michael Neely, and Xiaohan Wei. 2017. Online convex optimization with stochastic constraints. *Advances in Neural Information Processing Systems* 30 (2017).
- [47] Hao Yu and Michael J Neely. 2020. A low complexity algorithm with $O(\sqrt{T})$ regret and $O(1)$ constraint violations for online convex optimization with long term constraints. *Journal of Machine Learning Research* 21, 1 (2020), 1–24.
- [48] Congde Yuan, Mengzhuo Guo, Chaoneng Xiang, Shuangyang Wang, Guoqing Song, and Qingpeng Zhang. 2022. An actor-critic reinforcement learning model for optimal bidding in online display advertising. In *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*. 3604–3613.
- [49] Haoqi Zhang, Lvyin Niu, Zhenzhe Zheng, Zhilin Zhang, Shan Gu, Fan Wu, Chuan Yu, Jian Xu, Guihai Chen, and Bo Zheng. 2023. A personalized automated bidding framework for fairness-aware online advertising. In *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 5544–5553.
- [50] Weinan Zhang, Yifei Rong, Jun Wang, Tianchi Zhu, and Xiaofan Wang. 2016. Feedback control of real-time display advertising. In *Proceedings of the Ninth ACM International Conference on Web Search and Data Mining*. 407–416.
- [51] Jun Zhao, Guang Qiu, Ziyu Guan, Wei Zhao, and Xiaofei He. 2018. Deep reinforcement learning for sponsored search real-time bidding. In *Proceedings of the 24th ACM SIGKDD international conference on knowledge discovery & data mining*. 1021–1030.

A More details about AuctionNet benchmark

A.1 Offline Dataset

The AuctionNet dataset² [39] is collected from the logs of an advertising bidding environment, where multiple automated bidding agents compete for arriving impression opportunities. Each period is divided into 48 independent time steps, with approximately 480,000 impression opportunities per time step. The dataset consists of a dense subset and a sparse subset, which mainly differ in terms of conversion rates and CPA range. Notably, **the conversion rate of the sparse subset more closely reflects realistic scenarios**, whereas the conversion rate of the dense subset is approximately an order of magnitude higher than typically observed in practice. Key statistics of the dataset are summarized in Table 3.

Table 3: Data statistics.

Parameters	AuctionNet	AuctionNet-Sparse
Trajectories	479,376	479,376
Delivery Periods	9,987	9,987
Time steps in a trajectory	48	48
State dimension	16	16
Action dimension	1	1
Return-To-Go Dimension	1	1
Budget range	[450, 11850]	[2000, 4850]
Action range	[0, 493]	[0, 589]
Impression's value range	[0, 1]	[0, 1]
CPA range	[6, 12]	[60, 130]
Total conversion range	[0, 1512]	[0, 57]
Average conversion rate	0.0051	0.0005

In addition, for each subset, AuctionNet provides both impression-level training data and trajectory training data. The trajectory data is aggregated from the impression-level data, with the formats described as follows:

Description of Impression-level Training Data Format

- **deliveryPeriodIndex**: Represents the index of the current delivery period.
- **advertiserNumber**: Represents the unique identifier of the advertiser.
- **advertiserCategoryIndex**: Represents the index of the advertiser's industry category.
- **budget**: Represents the advertiser's budget for a delivery period.
- **CPAConstraint**: Represents the CPA constraint of the advertiser.
- **timeStepIndex**: Represents the index of the current decision time step.
- **remainingBudget**: Represents the advertiser's remaining budget before the current step.
- **pvIndex**: Represents the index of the impression opportunity.
- **pValue**: Represents the conversion action probability when the advertisement is exposed to the customer.
- **pValueSigma**: Represents the prediction probability uncertainty.
- **bid**: Represents the advertiser's bid for the impression opportunity.
- **xi**: Represents the winning status of the advertiser for the impression opportunity, where 1 implies winning the opportunity and 0 suggests not winning the opportunity.
- **adSlot**: Represents the won ad slot. The value ranges from 1 to 3, with 0 indicating not winning the opportunity.
- **cost**: Represents the cost that the advertiser needs to pay if the ad is exposed to the customer.
- **isExposed**: Represents whether the ad in the slot was displayed to the customer, where 1 implies the ad is exposed and 0 suggests not exposed.
- **conversionAction**: Represents whether the conversion action has occurred, where 1 implies the occurrence of the conversion action and 0 suggests that it has not occurred.
- **leastWinningCost**: Represents the minimum cost to win the impression opportunity, i.e., the 4th highest bid of the impression opportunity.
- **isEnd**: Represents the completion status of the advertising period, where 1 implies either the final decision step of the delivery period or the advertiser's remaining budget falling below the system-set minimum remaining budget.

Description of Trajectory Data Format

- **deliveryPeriodIndex**: Indicates the index of the current delivery period.
- **advertiserNumber**: The unique identifier for the advertiser.
- **advertiserCategoryIndex**: The industry category index of the advertiser.
- **budget**: The advertiser's budget for one delivery period.
- **CPAConstraint**: The advertiser's CPA (cost per action) constraint.
- **realAllCost**: The total cost incurred by the advertiser during the delivery period.
- **realAllConversion**: The total number of conversions generated by the advertiser during the delivery period.
- **timeStepIndex**: The index of the current decision step.
- **state**: The state at the current decision step.

²<https://github.com/alimama-tech/AuctionNet>

- **action**: The action taken at the current decision step.
- **reward**: The sparse reward at the current decision step (i.e., the sum of conversions in the current step).
- **reward_continuous**: The continuous reward at the current decision step (i.e., the sum of pValues for all impression traffic in the current step).
- **done**: Indicates whether the delivery period is completed; 1 denotes either the last step of the period or the remaining budget of the advertiser falls below the minimum threshold set by the system.
- **next_state**: The state at the next decision step.

A.2 Simulation Environment

AuctionNet [39] implements a standardized simulated online bidding environment. The bidding models to be evaluated are required to represent a specific advertiser and compete against baseline models provided by the environment, under given budget and CPA constraints. The simulation environment runs the bidding models multiple times with different advertiser configurations and various simulated traffic data.

Specifically, the bidding model operates on 48 advertisers, and each advertiser is evaluated over 7 rounds. Each advertiser bids independently and has no knowledge of other advertisers' bidding information. The simulated online bidding environment is a dynamic game environment, where the auction process is conducted in real time based on the bids from different models, whereas in the offline evaluation, the winner is determined only by the historical minimum winning price. This enables the simulation environment to provide more realistic testing than offline testing.

B Deploy Details of Online A/B Test

To evaluate the effectiveness of QGA, we implemented it on Taobao within the Multiple Constraint Bidding (MCB) framework. In this setting, advertisers specify a budget and may optionally impose Cost Per Click (CPC) or Return on Investment (ROI) constraints. The bidding strategy aims to maximize traffic value while satisfying

these requirements. For comparison, the control group adopted the original DT, with both groups employing identical intervention strategies except for the bidding policies. The experimental setup is outlined as follows:

- **Action**: To stabilize bidding outcomes, the bid coefficient λ is determined using a windowed average over the preceding two hours, which comprises E time steps, as follows:

$$\lambda_t = a_t + \frac{1}{|E|} \sum_{t'=t-E}^{t-1} \lambda_{t'}$$

where a_t represents the action output by QGA at time step t .

- **State**: A total of 19 state features are collected by recording the interactions between online ad bids and the platform's bidding environment:
 - **Campaign status**: Proportion of remaining bidding steps, proportion of remaining budget, and deviation from CPC.
 - **Historical and recent state averages (over the last three time steps)**: Number of impressions, number of clicks, number of conversions, advertising GMV (Gross Merchandise Value), click-through rate (CTR), conversion rate (CVR), ad cost, and bid.
- **Return-to-Go**: The reward function is defined as the expected GMV from the current time step to the end of the day, under the constraints of CPC and budget.

For training, we collected data from the most recent seven days, yielding approximately 130,000 trajectories, each comprising 48 time steps. Online A/B testing was performed over a consecutive ten-day period, which included seven regular days and three promotional days. For each MCB campaign, 33% of the product items were allocated to QGA. The proposed method consistently demonstrates strong adaptability during both regular and promotional periods, achieving notable improvements in Ad GMV (up to 4.70%) and ROI (up to 2.49%) with only a slight increase in Ad Cost. These results suggest that the method enables effective and flexible cost allocation, as well as improved identification of optimal bidding strategies in dynamic traffic environments.